Explore US Bikeshare Clip Navigator

You do not need to watch either complete recording. Choose the clip that matches your current blocker.

Disclaimer: The text in this document is AI-generated and should be reviewed before learner-facing publication.

Minimum Path

- New to the cohort? Watch C01.

- Starting the project? Watch C12 before coding.

- Stuck on Python syntax? Choose C02-C08 based on the blocker.

- Stuck on pandas/statistics? Choose C09-C11.

- Still blocked after trying? Post in the community or book a 1:1.

Clip Index

Clip	Use when	Time	Slides	Video
C01	Know how to approach the course, get help, and use the community instead of waiting passively.	5m 16s	4-7	C01-course-strategy-support.mp4
C02	Refresh variables, data types, operators, print output, and the first basics quiz.	10m 32s	10-11	C02-python-variables-operators.mp4
C03	Understand lists, tuples, dictionaries, and sets well enough to read the project constants and data choices.	6m 03s	12	C03-data-structures-project-dictionaries.mp4
C04	Use if/elif/else and loops, then understand the truthiness trap that affects user input checks.	7m 34s	13-14	C04-control-flow-truthiness.mp4
C05	Organize reusable project code with functions and understand where variables live.	8m 35s	16-17	C05-functions-scope.mp4
C06	Recognize lambda expressions and understand the late-binding quiz without treating it as required project syntax.	8m 53s	18-19	C06-lambda-late-binding.mp4
C07	Run Python scripts, import local code/modules, install packages, and read simple error handling.	5m 58s	22-25	C07-scripts-imports-errors.mp4
C08	Debug the Bikeshare project by printing values or using `pdb` breakpoints.	7m 25s	26-27	C08-debugging-pdb.mp4
C09	Understand why NumPy exists and how pandas builds a table/column model for data analysis.	6m 04s	30-31	C09-numpy-pandas-mental-model.mp4
C10	Load CSV data, select columns/rows, filter values, and use docs or AI when pandas syntax is unfamiliar.	9m 43s	31-33	C10-pandas-filtering-docs.mp4
C11	Use `.info()`, `.describe()`, `.value_counts()`, and grouping/aggregation for project-style statistics.	6m 53s	34-38	C11-pandas-summary-grouping.mp4
C12	Understand what the Explore US Bikeshare project must do and where to go when blocked.	3m 05s	39-41	C12-project-requirements-next-steps.mp4

Learner Guide

C01-course-strategy-support: Course strategy and support path

Use this clip when:
- You are new to the cohort and want to know how to get help.
- You are deciding whether to start with lessons or the project.

After watching, you should be able to:
- Use the community space for questions that can help other learners too.
- Follow the project-first learning strategy: work until stuck, then return to lessons.
- Understand that the course exercises are useful practice, but the projects are the completion requirement.

You can skip this if:
- You already know where sessions, recordings, community, and 1:1 support live.

Deck slides: 4-7 | Time: 5 min 16 sec | Zoom clip: C01 | Video file: C01-course-strategy-support.mp4 | Captions: C01-course-strategy-support.vtt

C02-python-variables-operators: Python basics: variables, types, operators, and print

Use this clip when:
- You need a quick Python refresher before reading or editing the Bikeshare script.
- You are unsure what variables, booleans, type conversion, or print output mean.

After watching, you should be able to:
- Explain how a value is stored in a variable.
- Recognize primitive and complex data types.
- Understand why a == 3 prints True in the quiz.
- Use a notebook/workspace to test small expressions before writing project code.

You can skip this if:
- You are already comfortable reading simple Python expressions.

Deck slides: 10-11 | Time: 10 min 32 sec | Zoom clip: C02 | Video file: C02-python-variables-operators.mp4 | Captions: C02-python-variables-operators.vtt

C03-data-structures-project-dictionaries: Data structures for the Bikeshare project

Use this clip when:
- You see lists, tuples, dictionaries, or sets in starter code and want the short version.
- You want to understand why CITY_DATA works like a lookup table.

After watching, you should be able to:
- Distinguish list, tuple, dictionary, and set use cases.
- Read dictionary syntax as key-value pairs.
- Explain why dictionaries are useful for mapping city names to CSV files.

You can skip this if:
- You can already read and modify Python containers comfortably.

Deck slides: 12 | Time: 6 min 03 sec | Zoom clip: C03 | Video file: C03-data-structures-project-dictionaries.mp4 | Captions: C03-data-structures-project-dictionaries.vtt

C04-control-flow-truthiness: Control flow and truthy values

Use this clip when:
- Your script needs to choose a path based on city, month, day, or restart input.
- You are confused by why if city: runs before elif city == "Chicago".

After watching, you should be able to:
- Read basic if, elif, and else branches.
- Read a simple loop over city values.
- Explain why non-empty strings are treated as true in Python.
- Avoid placing a broad truthiness check before a more specific comparison.

You can skip this if:
- You are already confident with conditionals and truthy/falsy values.

Deck slides: 13-14 | Time: 7 min 34 sec | Zoom clip: C04 | Video file: C04-control-flow-truthiness.mp4 | Captions: C04-control-flow-truthiness.vtt

C05-functions-scope: Functions and variable scope

Use this clip when:
- You are turning repeated code into functions.
- You are unsure what arguments, return values, docstrings, or variable scope mean.

After watching, you should be able to:
- Explain why functions make project code easier to reuse and read.
- Identify function definitions, arguments, docstrings, function body, return values, and calls.
- Understand why a variable created inside a function may not exist outside it.

You can skip this if:
- Your project functions are already organized and readable.

Deck slides: 16-17 | Time: 8 min 35 sec | Zoom clip: C05 | Video file: C05-functions-scope.mp4 | Captions: C05-functions-scope.vtt

C06-lambda-late-binding: Lambda expressions and the late-binding trap

Use this clip when:
- You saw a lambda expression and want to know what it means.
- You want to understand why the quiz prints [3, 3, 3].

After watching, you should be able to:
- Describe lambda as a short anonymous function.
- Recognize when a function can be passed into another function such as map.
- Explain the late-binding trap: the lambdas reference x, and by the time they run, x is 3.

You can skip this if:
- You only want the minimum needed to complete Explore US Bikeshare.

Deck slides: 18-19 | Time: 8 min 53 sec | Zoom clip: C06 | Video file: C06-lambda-late-binding.mp4 | Captions: C06-lambda-late-binding.vtt

C07-scripts-imports-errors: Running scripts, imports, modules, and errors

Use this clip when:
- You are moving from notebook cells to a .py project script.
- You need to understand imports, package installation, or try/except blocks.

After watching, you should be able to:
- Run a Python script from a terminal.
- Recognize imports from local files and external modules.
- Understand why import pandas as pd creates the pd alias.
- Read basic try and except paths for invalid user input.

You can skip this if:
- You already know how to run the project script and handle simple errors.

Deck slides: 22-25 | Time: 5 min 58 sec | Zoom clip: C07 | Video file: C07-scripts-imports-errors.mp4 | Captions: C07-scripts-imports-errors.vtt

C08-debugging-pdb: Debugging the project with print checks and pdb

Use this clip when:
- Your script runs but the result is wrong.
- You want to inspect variable values while the program is running.

After watching, you should be able to:
- Treat errors as useful clues, not failure.
- Use print checks to discover which path your code takes.
- Understand what a pdb.set_trace() breakpoint does.
- Inspect variables such as the most common month inside the running project.

You can skip this if:
- Your current blocker is about pandas syntax rather than debugging flow.

Deck slides: 26-27 | Time: 7 min 25 sec | Zoom clip: C08 | Video file: C08-debugging-pdb.mp4 | Captions: C08-debugging-pdb.vtt

C09-numpy-pandas-mental-model: NumPy and pandas mental model

Use this clip when:
- You want the big picture before using pandas.
- You are unsure why NumPy is mentioned if the project mostly uses pandas.

After watching, you should be able to:
- Explain that NumPy helps Python do fast numerical work.
- Recognize arrays, broadcasting, and normalization at a high level.
- Understand that pandas gives you DataFrames and Series for table-like data.

You can skip this if:
- You only need project-specific pandas commands right now.

Deck slides: 30-31 | Time: 6 min 04 sec | Zoom clip: C09 | Video file: C09-numpy-pandas-mental-model.mp4 | Captions: C09-numpy-pandas-mental-model.vtt

C10-pandas-filtering-docs: pandas essentials: reading, selecting, and filtering

Use this clip when:
- You need pandas basics for the Bikeshare statistics.
- You are unsure how DataFrames, Series, column selection, masks, or .loc work.

After watching, you should be able to:
- Describe a DataFrame as a table and a Series as a column.
- Read a CSV into pandas.
- Select a column and filter rows with a boolean mask.
- Look up pandas syntax instead of trying to memorize every function.

You can skip this if:
- You are already comfortable with pandas selection and filtering.

Deck slides: 31-33 | Time: 9 min 43 sec | Zoom clip: C10 | Video file: C10-pandas-filtering-docs.mp4 | Captions: C10-pandas-filtering-docs.vtt

C11-pandas-summary-grouping: pandas statistics: summary functions and grouping

Use this clip when:
- You need to compute or inspect project statistics.
- You are deciding which pandas function helps answer a data question.

After watching, you should be able to:
- Use .info() to inspect columns and data types.
- Use .describe() for numeric summaries.
- Use .value_counts() for category counts.
- Recognize groupby and aggregation as the pandas version of pivot-style summaries.
- Use an AI assistant or docs to recover syntax while still understanding the goal.

You can skip this if:
- You are not yet working on the statistics portion of the project.

Deck slides: 34-38 | Time: 6 min 53 sec | Zoom clip: C11 | Video file: C11-pandas-summary-grouping.mp4 | Captions: C11-pandas-summary-grouping.vtt

C12-project-requirements-next-steps: Project requirements and next steps

Use this clip when:
- You are about to start the final Bikeshare project.
- You want the short checklist of what the script must do.

After watching, you should be able to:
- Name the required statistics categories: time, station, trip duration, and user-related statistics.
- Remember to print five rows of raw data at a time.
- Remember to ask whether the user wants to restart.
- Use practice problems, community posts, AI help, or a 1:1 session when blocked.

You can skip this if:
- You already have the rubric open and only need help with a specific function.

Deck slides: 39-41 | Time: 3 min 05 sec | Zoom clip: C12 | Video file: C12-project-requirements-next-steps.mp4 | Captions: C12-project-requirements-next-steps.vtt